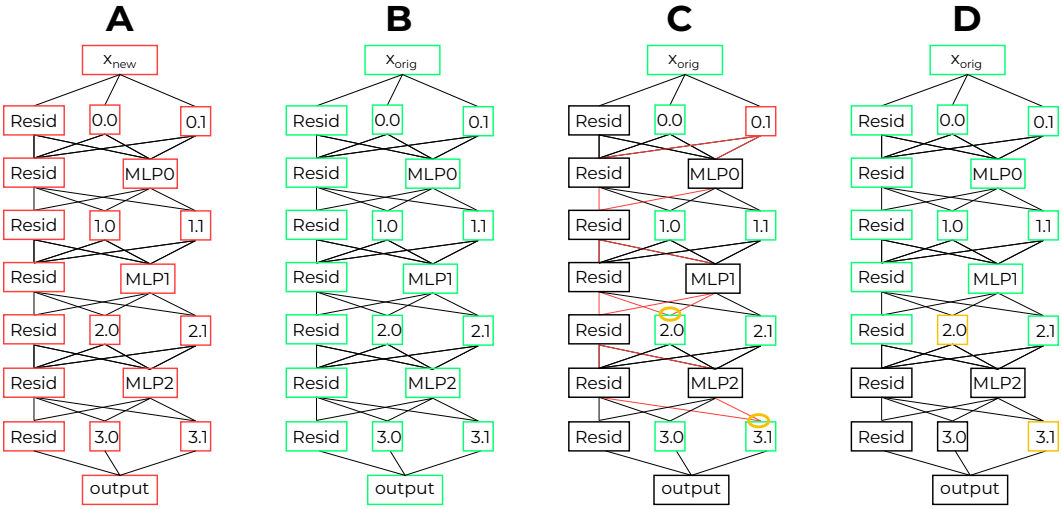


Figure 11: Illustration of the four forward passes in the path patching method. $h = 0.1$ and $R = \{2.0, 3.1\}$. The layer norms are not shown for clarity. In the forward pass C, we show in red all the paths through which h can influence the value of heads in R. Nodes in black are recomputed, nodes in color are patched or frozen to their corresponding values.

between S1 and S2 (such pointer arithmetic behavior has already been observed in Olsson et al. (2022)). When patching in S2 Inhibition outputs from a distribution where prefixes to sentences are longer (but the distance between S1 and S2 is constant), the logit difference doesn't change (3.56 before patching vs 3.57 after). This suggests that the positional signal doesn't depend on the absolute position of the tokens, as long as the relative position of S1 and S2 stays the same.

B Path patching

In this section we describe the technical details of the path patching technique we used in many experiments. This technique depends on

- x_{orig} the original data point,
- x_{new} the new data point,
- h the *sender* attention head, and
- $R \subseteq M$ the set of *receiver* nodes in the model's computational graph M. In our case, R is either the input (key, query or values) of a set of attention heads or the end state of the residual stream[3].

It is then implemented as Algorithm 1, which can be summarized in five steps.

1. Gather activations on x_{orig} and x_{new}. [Lines 1-2, Figure 11A-B]
2. Freeze[4] all the heads to their activation on x_{orig} except h that is patched to its activation on x_{new}. [Lines 4-5, Figure 11C]
3. Run a forward pass of the model on x_{orig} with the frozen and patched nodes (MLPs and layer norm are recomputed). [Line 7, Figure 11C]
4. In this forward pass, save the activation of the model components $r \in R$ as if they were recomputed (but this value is overwritten in the forward pass because they are frozen). [Line 6-9, Figure 11C]

[3]The model uses the end state of the residual stream to compute the logits (after a layer norm), so we choose R equal to this end state when we wish to measure the direct effect of h on the logits.

[4]We use 'freezing' of an activation a to refer to the special case of patching when we begin a forward pass on an input x, edit some earlier activations, and then want to patch a to its value on a forward pass of x (which did not have the edits of earlier activations).

5. Run a last forward pass on x_{orig} patching the receiver nodes in R to the saved values. [Lines 12-20, Figure 11D]

Each of the A variables in Algorithm 1 (A_{orig}, A_{new} and A_{patch}) stores activations for all nodes in M. A_{orig} and A_{new} contain output activations while A_{patch} contains the activation of the nodes in R (keys, queries or values of attention heads, or the end state of the residual stream).

In forward pass C of the algorithm all attention heads except h have their output frozen to their value on x_{orig}. This means that we remove the influence of every path including one (or more) intermediate attention heads p of the form $h \to p \to r$, where $r \in R$. In all cases in the paper, we measure the logit difference on the forward pass D compared to the logit difference of the model, as a measure of the importance of the path $h \to r$. We additionally compute this logit difference as an average over $N > 200$ pairs $(x_{\text{orig}}, x_{\text{new}})$.

Algorithm 1 Path patching

1: Compute all activations A_{new} on x_{new} (forward pass A)
2: Compute all activations A_{orig} on x_{orig} (forward pass B)
3: **for** $r \in R$ **do**
4: $\quad A^r_{\text{patch}} \leftarrow A_{\text{orig}}$
5: $\quad A^r_{\text{patch}}[h] \leftarrow A^r_{\text{new}}[h]$
6: $\quad$ **for** each MLP m between h and r **do**
7: $\quad\quad$ Recompute $A^r_{\text{patch}}[m]$ from the activations in A^r_{patch} (forward pass C)
8: $\quad$ **end for**
9: $\quad$ Recompute $A^r_{\text{patch}}[r]$ from the activations in A^r_{patch} (forward pass C)
10: **end for**
11:
12: $A_{\text{final}} \leftarrow \emptyset$
13: **for** $c \in M$ nodes of the computational graph, in topologically sorted order, **do** (forward pass D)
14: $\quad$ **if** $c \in R$ **then**
15: $\quad\quad A_{\text{final}}[c] \leftarrow A^c_{\text{patch}}[c]$
16: $\quad$ **else**
17: $\quad\quad$ Compute $A_{\text{final}}[c]$ from activations in A_{final}
18: $\quad$ **end if**
19: **end for**
20: **return** $A_{\text{final}}[\text{Logits}]$

C Direct effect on S-Inhibition Heads' keys

In this section, we present the direct effect analysis of the S-Inhibition Heads' keys. The experiment is similar to the investigation of the S-Inhibition Heads' values presented in Section 3.2.

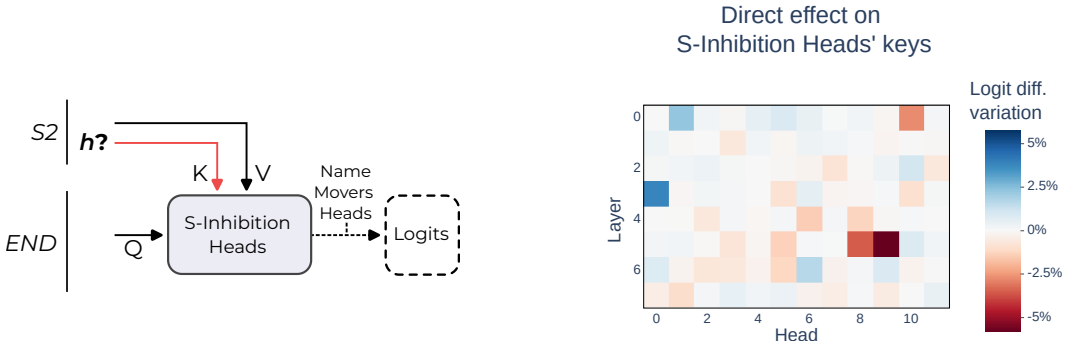
The results presented in Figure 12b show that some heads significantly influence the logit difference through S-Inhibition Heads' keys. We observe that Duplicate Token Heads (3.0 and 0.1) appear to also influence S-Inhibition Heads' values, but their effect is reversed.

Moreover, we identify 3 new heads influencing positively the logit difference: 5.9, 5.8 and 0.10.

Fuzzy Duplicate Token Head By looking at its attention pattern, we identified that 0.10 was paying attention to S1 from S2. However, the attention pattern was fuzzy, as intermediate tokens also have non-negligible attention probability. We thus call it a fuzzy Duplicate Token Head. On Open Web Text (OWT), the fuzzy Duplicate Token Head attend to duplicates in a short range (see Appendix I).

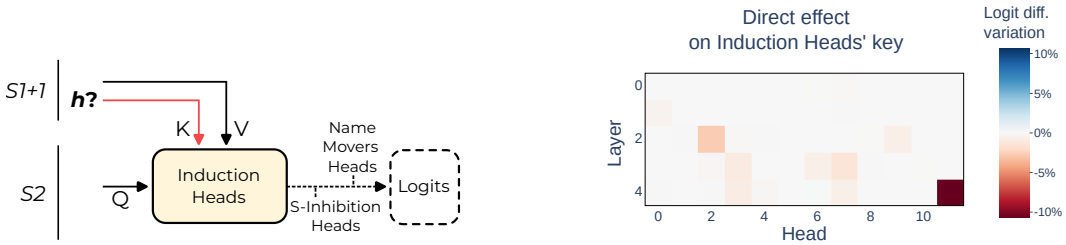
Fuzzy Induction Heads The heads 5.9, 5.8 are paying attention to S1+1. For 5.8, S1+1 is the token with the highest attention probability after the start token. But the absolute value is small (less than 0.1). The head 5.9 is paying attention to S1+1 but also to tokens before S. Because of these less interpretable attention patterns, we called them fuzzy Induction Heads. (see Appendix H for more details about their behavior outside IOI).

By influencing the keys of the S-Inhibition Heads, we hypothesize that those heads are amplifying the positional signal written by the other Induction and Duplicate Token Heads.



(a) (b)

Figure 12: (a) Diagram of the direct effect experiment on S-Inhibition Heads' keys at S2. (b) Result of the path patching experiments for direct effect experiment on S-Inhibition Heads' keys.



(a) (b)

Figure 13: (a) Diagram of the direct effect experiment on Induction Heads' keys at S1+1. The effects on logits are mediated by S-Inhibition Heads and Name Mover Heads (b) Result of the path patching experiments on Induction Heads' keys at S1+1.

D Identification of Previous Token Heads

Induction Heads rely on key composition with Previous Token Heads to recognize patterns of the form `[A] [B] ... [A]`. In the context of IOI, the repeated token `[A]` is S2. We thus searched for heads directly affecting Induction Heads keys at the S1+1 position (the `[B]` token in the general pattern). For this, we used path patching. The results of the experiment are visible in Figure 13. We identify two main heads causing a decrease in logit difference (and thus contributing positively to the logit difference): 4.11 and 2.2. These heads pay primary attention to the previous token. This is coherent with the Previous Token Heads we were expecting. We thoroughly investigate their attention patterns outside of p_{IOI} in Appendix H.

Olsson et al. (2022) also describes an induction mechanism relying on query composition in GPT-2. We performed path patching to investigate heads influencing the Induction Heads queries at the S2 position, but did not find any significant effect.

E IOI Templates

We list all the templates we used in Figure 14. Each name was drawn from a list of 100 English first names, while the place and the object were chosen among a hand-made list of 20 common words. All the words chosen were one token long to ensure proper sequence alignment computation of the mean activations.